

Boolean Function/ Logic Function

a) Boolean function represented as an algebraic expression.

i. $F_1 = \overline{xyz}$

ii. $F_2 = \overline{x+y}z$

iii. $F_3 = x'y'z + x'yz + xy'$

iv. $F_4 = xy' + xz$

b) Boolean function represented in truth table.

X	Y	Z	F1	F2	F3	F4
0	0	0	0			
0	0	1	0	1		
0	1	0	0			
0	1	1	0			
1	0	0	0	1		
1	0	1	0			
1	1	0	1			
1	1	1	0			

c) Boolean function represented in logic diagram. **(Implementation of Boolean Function using gates)**

Question:

Draw a logic gate that implements the following:

a) $F = AB + CD' + B'C$

b) $F = (A+B)(B'+C)(C'+D+E)$

Boolean's Laws and Theorems:

Boolean's Algebra is used to analyze and simplify the digital circuits. Because it used only two binary numbers, it is also called binary algebra or logical algebra. It is also called binary algebra or logical algebra.

The rules of Boolean's Algebra are different from those of the conventional algebra as:

- i) Symbol used in Boolean Algebra (usually letters) do not represent numerical values.
- ii) Arithmetic operation like subtraction, division are not performed. Also, there are no fractions, squares, square root, logarithm and imaginary numbers.
- iii) Boolean Algebra allows only two possible values i.e. 1 and 0 for any variable.

The Boolean Laws are: (Basic Theorem of Boolean Algebra)

1. Commutative Law:

This Law allows change in the position of the input variable in OR and AND expression.

- i) $A+B=B+A$
- ii) $A.B=B.A$

2. Associative Law:

The law states that it make no difference in what order the variables are grouped when ORing or ANDing more than two variables.

- i) $A+(B+C)=(A+B)+C$
- ii) $A.(B.C)=(A.B).C$

3. Distributive Law:

This law permits factoring or multiplying out an expression.

- i) $A(B+C)=AB+AC$
- ii) $AB+AC=A(B+C)$
- iii) $A+(BC)=(A+B).(A+C)$

4. Inversion Law/Complementary Law:

- i) $A=A''$

5. Identity law:

- i) $A+0=A$
- ii) $A.1=A$

6. Complement Law:

- i) $A+A'=1$
- ii) $A.A'=0$

7. OR Law:

- i) $A+0=A$
- ii) $A+1=1$
- iii) $A+A=A$
- iv) $A+A'=1$
- v) $A+AB=A$
- vi) $A+A'B=A+B$
- vii) $(A+B)(A+C)=(A+BC)$

8. AND Law:

- i) $A.0=0$
- ii) $A.1=A$
- iii) $A.A=A$
- iv) $A.A'=0$

9. Duality Theorem:

It states that “**Every algebraic expression deducible from the postulates of Boolean algebra remains valid if the operators and identity elements are interchanged**”. In a two-valued Boolean algebra, the identity elements and the elements of the set B are the same: 1 and 0. If the **dual** of an algebraic expression is desired, we simply interchange OR and AND operators and replace 1's by 0's and 0's by 1's.

E.g. $X \cdot Y + Z' = (X' + Y') \cdot Z$

OR

It states that each Boolean expression has its dual which is as true as original expression.

Example:

$$XY' + XYZ + YZ' = (X + Y')(X + Y + Z)(Y + Z')$$

The dual of given Boolean expression can be obtained by following the procedure given below.

- i) Changing each OR(+) sign to AND(.) sign
- ii) Changing each AND(.) sign to OR(+) sign
- iii) Complementing any 0 or 1 appearing in the expression.

For example:

a) For the expression $A+0=A$, the dual relation is $A.1=A$

i.e. $A+0=A \implies A.1=A$

a) $A+1=1 \implies A.0=0$

b) $A+A=A \implies A.A=A$

c) $A(B+C) \implies AB+AC$

d) By using each OR and AND operation, we get the dual relation.

$$A+B.C=(A+B).(A+C)$$

e) $A+A'B=AB \implies A.(A'+B)=A+B$

f) $A(B+C) \implies A+(B.C)$

Que. Simplify using Boolean Algebra:

- i)
$$\begin{aligned} & . x(x' + y) \\ & = xx' + xy \\ & = 0 + xy \\ & = xy \end{aligned}$$
- ii)
$$\begin{aligned} & x + x'y \\ & = (x + x')(x + y) \\ & = 1(x + y) \end{aligned}$$

$$= x + y$$

$$\begin{aligned} \text{iii)} \quad & (x + y)(x + y') \\ &= x + xy + xy' + yy' \\ &= (1 + y + y') \\ &= x \end{aligned}$$

Que. Prove that:

- i.
- ii. $(x + y)(x' + y) = y$

Proof	Identity Name
$(x+y)(\bar{x}+y) = x\bar{x}+xy+y\bar{x}+yy$	Distributive Law
$= 0+xy+y\bar{x}+yy$	Inverse Law
$= 0+xy+y\bar{x}+y$	Idempotent Law
$= xy+y\bar{x}+y$	Identity Law
$= y(x+\bar{x})+y$	Distributive Law (and Commutative Law)
$= y(1)+y$	Inverse Law
$= y+y$	Identity Law
$= y$	Idempotent Law

- iii. Prove the Boolean expression: $AB + AB'C + A'BC = AB + AC + BC$

Soln:

$$\begin{aligned} & AB + AB'C + A'BC \\ &= A(B + B'C) + A'BC \\ &= A(B + C) + A'BC \quad [\because A + A'B = A + B] \\ &= AB + AC + A'BC \\ &= AB + (A + A'B)C \\ &= AB + (A + B)C \\ &= AB + AC + BC \end{aligned}$$

Questions:

1. Minimize the expression: $AB + A\bar{C} + BC$ using theorem and properties of Boolean algebra.

2. Simplify $X'Y'Z' + X'YZ + XY'$ using Boolean algebra.

3. Solve:

a) $A.B.C + A.B.C' + A'.B.C + A'B$

b) $X'Y'Z' + X'YZ + XY'$

c) $XY + X'Z + YZ$

4. Prove that:

a) $A + A'B = A + B$

b) $A + AB = A$

c) $(A+B)(A+C) = A+BC$

d) $X'Y'Z' + X'YZ' + XYZ' + XY'Z' = Z'$

e) $A.B' + B'C + A'C = AB' + A'C$

f) $A(A'+C)(A'B+C)(A'BC+C') = 0$